

Random Proposal Normalization Loss (RPNL)

Problem: Standard depth losses normalize globally. Close objects dominate; distant detail is lost.

Analogy from course: This is the same problem *adaptive histogram equalization* solves for image contrast. Global normalization washes out local structure.

RPNL solution:

1. Randomly crop K patches from the depth map
2. Normalize depth within each patch independently
3. Compute loss per patch, then average

This forces the network to preserve *local*

